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Ether chain modified electrochromic polymer based on EDOT and benzene

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ABSTRACT

Polar side chain plays an critical role in regulating molecular aggregation, film morphology and electrochromic properties of conjugated polymer. Here an ether chain modified conjugated polymer poly(EB-ether) based on EDOT and benzene was prepared at low oxidation potential by electrochemical method in EtOH-CH_2Cl_2 (v/v = 5/1) containing 0.1 M Bu_4NPF_6. The prepared poly(EB-ether) could achieve similar electrochromic parameters with alkoxy chain modified poly(EB-alkoxy), such as optical contrast of 35 %, response time of 0.8 s, coloration efficiency of 283 C^{-1} cm^2, and reversible color change between dark red and blue. Electrochromic results indicate that high-performance poly(EB-ether) film could be prepared in mixed solvent containing little CH_2Cl_2. Therefore, the introduction of ether chain in precursor is a feasible method to decrease or avoid the use of halogenated solvent in electrochemical polymerization process of precursor.

1. Introduction

Nowadays the electronic products based on electrochromism have been pursued by the society, which own increasing market share in the fields of architectural glass, flexible display, camouflage material, etc [1–5]. Conjugated polymers as electrochromic materials exhibit wide application prospect because of their flexibility, rich color, fast response time, easy preparation, etc [6–9]. To meet the property requirement of constantly updated electronic products, the high-performance electrochromic conjugated polymer is urgently needed.

These strategies to improve the properties of conjugated polymers include modifying conjugated block, changing side chain structure, post-treating polymer film, etc. [10–18]. Among these strategies, changing side chain structure possesses advantage in optimizing electrochromic properties under condition of remaining unchanged absorption region, such as response time, optical contrast, cycle stability, etc. For example, the cross-linked group was introduced into side chain of polythiophene derivative by Ming et al, which exhibits higher optical contrast and better cycle stability than referenced polymer with different side chain [19]. Ether chain as polar functional group plays important role on the optimization of polymer performance due to the modification of polymer backbone with polar group to assist in penetration of inorganic electrolyte ions. For example, Reynolds group reported an oligo(ether) functionalized dioxythiophene copolymer, which processes good electroactivity in aqueous and organic electrolyte systems. Also, the copolymer exhibits rapid response time (<2 s) and high optical contrast (>70 %) due to the polar group oligo(ether) assisting in the transport of doping ions [20]. Besides, the introduction of polar chain makes precursor polymerize in green solvent, such as solvent alcohol and H_2O. In general, the halogenated solvents (such as CH_2Cl_2) are used in electrochemical polymerization of precursor in previous reports [21,22]. It is worth mentioning that halogenated solvents are harmful to environment and people's health. Therefore, the introduction of ether chain not only optimizes electrochromic properties of polymer film, but also modulates polymerization system of precursor.

Previous studies demonstrate that the copolymers containing thiophene and benzene exhibit good electrochromic properties, such as fast response time, high coloration efficiency, etc. [23–28]. For example, Reynolds group reported a series of high-bandgap yellow electrochromic polymers based on thiophene-benzene derivatives, which show high contrast of >65 % and fast response time [29]. Also, the benzene ring possesses two or more substitution sites, which provides opportunity to modify the properties of corresponding polymers by introducing functional group. For example, Xu group prepared a new copolymer based on bis(alkoxy) *ortho*-substituted benzene and thiophene derivative through changing substitution position of bis(alkoxy), which exhibit better cycle stability and richer colors than reference polymer [30]. It is worth mentioning that 3,4-ethoxylenedioxythiophene (EDOT) as a star

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molecule in thiophene derivatives is often used to construct conjugated polymer applied in different fields [31–33]. And, many conjugated polymers containing EDOT exhibit good electrochromic properties, such as the unique color change of green-transparent and black-transparent.

In this paper, the benzene derivative and EDOT based precursor 5,5′-(2,5-bis(2-ethoxyethoxy)-1,4-phenylene)bis(2,3-dihydrothieno[3,4-b][1,4]dioxine) (EB-ether) with ether chains was prepared, and electrochemical polymerized to corresponding polymer poly(EB-ether) in EtOH-CH_2Cl_2 (v/v = 5/1) containing 0.1 M Bu_4NPF_6 (Scheme 1). The electrochemical and electrochromic properties of poly(EB-ether) in H_2O-KCl (1.0 mol/L) were investigated systematically. Poly(EB-ether) could achieve optical contrast (ΔT) of 35 %, response time of 0.8 s, coloration efficiency (*CE*) of 283 C^{-1} cm^2, and reversible color change between dark red and blue. Finally, the electrochromic device based on poly(EB-ether) was constructed, which exhibit obvious color change and good stability.

2. Results and discussion

2.1. Synthesis of EB-ether

The synthetic route of EB-ether is shown in Scheme 1. The intermediate 1,4-dibromo-2,5-bis(2-ethoxyethoxy)benzene (BE-2Br) was synthesized by substitution reaction in the yield of 81 %. The target molecule EB-ether was prepared by Stille coupling reaction of BE-2Br and tributyl(2,3-dihydrothieno[3,4-*b*] [1,4]dioxin-5-yl)stannane (EDOT-Sn) in the yield of 72 %. The molecular structure of EB-ether was characterized by nuclear magnetic resonance (NMR), as followings: ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.72 (s, 2H), 6.37 (s, 2H), 4.31–4.30 (m, 4H), 4.25–4.24 (m, 4H), 4.20 (t, 4H), 3.88 (t, 4H), 3.65–3.60 (m, 4H), 1.25 (t, 6H). The synthesis details of EB-ether are present in Supporting Information.

2.2. Optical and electrochemical property of EB-ether

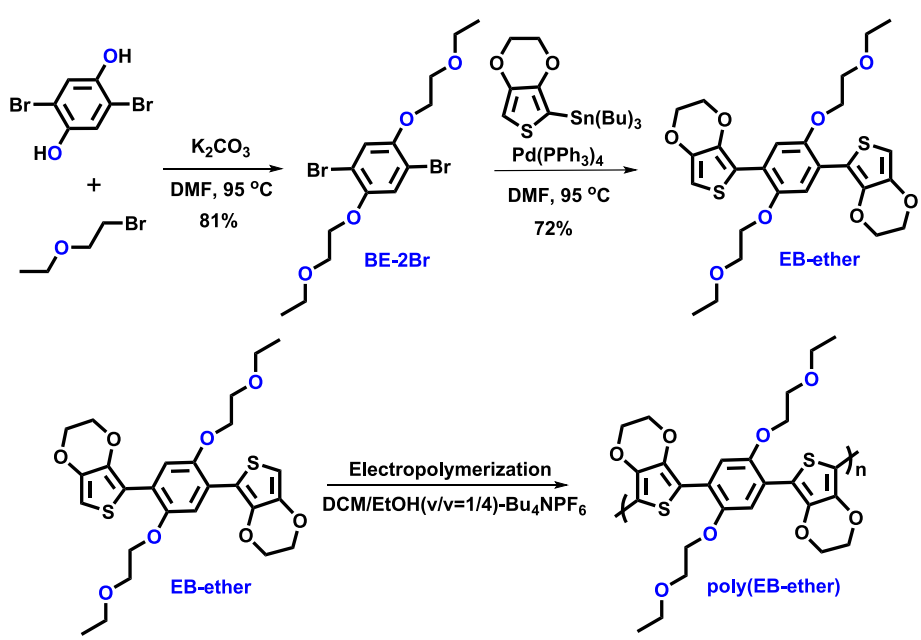
To explore the influence of ether side chain on optical absorption region, the absorption spectra of EB-ether and 5,5′-(2,5-bis(pentyloxy)-1,4-phenylene)bis(2,3-dihydrothieno[3,4-*b*] [1,4]dioxine) (EB-alkoxy) in CH_2Cl_2 were measured comparatively, as shown in Fig. 1a. EB-ether possesses optical absorption peaks at 308 nm, 320 nm and 358 nm, which is similar with EB-alkoxy (308 nm, 320 nm and 362 nm). The result indicates that the ether chain and alkoxy chain play similar role on optical absorption region of conjugated polymer.

Considering that EB-ether possesses polar side chain, EtOH-CH_2Cl_2 (v/v = 5/1) mixed solvent containing 0.1 M Bu_4NPF_6 was used to be electrolyte solution. Firstly, the onset oxidation potential of EB-ether was characterized by linear sweep voltammetry (LSV). It could be found that EB-ether exhibits low onset oxidation potential of 0.57 V (Fig. 1b), which is favor to prepare high-quality polymer film. Also, the electrochemical polymerization of EB-ether was performed by cyclic voltammetry (CV) in the potential window from −0.3 V to 0.8 V (Fig. 1b). As the cycle growing, the redox peak current densities of poly(EB-ether) increase gradually. Also, it could be observed that the blue polymer film was formed on the surface of working electrode. These results indicate that poly(EB-ether) film could be prepared in EtOH-CH_2Cl_2 (v/v = 5/1) mixed solvent containing 0.1 M Bu_4NPF_6, which prove that the introduction of ether chain in precursor is a feasible method to decrease or avoid the use of halogenated solvent in electrochemical polymerization of precursor.

2.3. FT-IR characterization and microstructure of polymer

To prove the successful synthesis of poly(EB-ether), fourier transform infrared spectoscopy (FT-IR) characterization of EB-ether and poly(EB-ether) were carried out, as shown in Fig. S3. And, the peak assignments of FT-IR spectra were summarized in Table S1. Firstly, the FT-IR spectrum of EB-ether is analyzed. The peak at 3104 cm^{-1} is ascribed to =C–H vibration of thiophene ring, which disappear in poly(EB-ether). The result indicates that the polymerization of precursor occurs at α position of thiophene ring. And, the other peaks of EB-ether are observed in poly(EB-ether), indicating the molecular structure isn't destroyed in polymerization process. It could be found that poly(EB-ether) exhibits wider peak than EB-ether, which due to the wide distribution of chain length.

The effective molecular aggregation is related to charge transport of conjugated polymer, which further affect its electrochromic properties. Most conjugated polymers obtained from electrochemcial method process amorphous forms not crystalline, resulting in that no sufficient information of molecular aggregation in X-ray diffraction spectrum could be obtained. Therefore, the molecular stacking of poly(EB-ether) was



Scheme 1. The synthetic routes of EB-ether and poly(EB-ether).

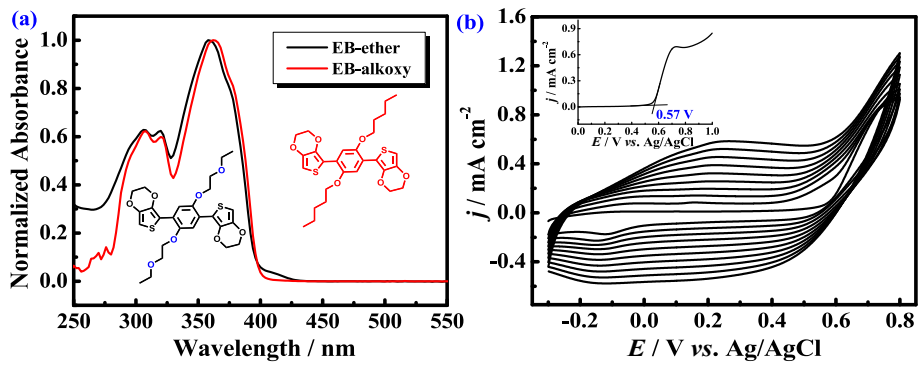


Fig. 1. (a) Optical absorption spectra of EB-ether and EB-alkoxy in CH_2Cl_2; (b) cyclic voltammogram of EB-ether in EtOH-CH_2Cl_2 (v/v = 5/1) containing 0.1 M Bu_4NPF_6 at scan rate of 100 mV s^{-1}, monomer concentration: 2 g/L, insert figure: LSV curve of EB-ether at scan rate of 50 mV s^{-1}.

calculated by density functional theory (DFT) on Vienna Ab initio Simulation Package (VASP), as shown in Fig. 2a. According to optimized molecular model, poly(EB-ether) possesses planar conjugated block, which is benefited to the formation of effective molecular aggregation. According to DFT result, the stacking distance of polymer chains was calculated to be 3.98 Å.

The morphology of poly(EB-ether) film was characterized by scanning electron microscope (SEM). The poly(EB-ether) film was prepared by potentiostatic method under potential of 1.0 V for 5 s. At the magnification of ×10000, the sparse ridge-like morphology of poly(EB-ether) is presented (Fig. 2b). When the magnification increases to ×200000, the porous structure with pore size of 50 ~ 100 nm could be observed on poly(EB-ether) film (Fig. 2c). This porous morphology is helpful to ion migrating in/out polymer film, which could improve the electrochromic properties of polymer, such as response time and coloration efficiency.

2.4. Electrochemical property of polymer

The electrochromic property of conjugated polymer is closely related to its electrochemical behaviour. Therefore, the electrochemical behavior of poly(EB-ether) was measured firstly by CV in monomer-free H_2O-KCl (1.0 mol/L) (Fig. 3a). The potential window is 0 V ~ 0.7 V. It could be found that there is linear relationship between the current densities of redox peaks and the scan rates (Fig. 3b), which implies the redox process of poly(EB-ether) is non-diffusion controlled. During redox process, poly(EB-ether) coated on working electrode exhibits obvious reversible color change, which indicates poly(EB-ether) prepared in EtOH-CH_2Cl_2 (v/v = 5/1) mixed solvent possesses good electrochemical property.

2.5. Electrochromic properties of polymer

Firstly, the spectroelectrochemical property of poly(EB-ether) was carried out in H_2O-KCl (1.0 mol/L) in colorimetric dish containing poly (EB-ether) modified ITO as working electrode, a Pt wire as counter electrode and an Ag/AgCl as reference electrode. The optical absorption spectra of poly(EB-ether) under different applied potentials were recorded, as shown in Fig. 4. At the neutral state, poly(EB-ether) exhibits an absorption peak at 489 nm, which is ascribed to π-π* transition of conjugated chain. It could be observed that neutral poly(EB-ether) is dark red. Also, no defined shape of absorption peak is observed for poly (EB-ether) (Fig. 4a), which is different from poly(EB-alkoxy). Previous reports demonstrate that defined shape of absorption peak is ascribed to strong molecular aggregation [34]. The phenomenon indicates that ether chain could modify molecular aggregation of conjugated polymer.

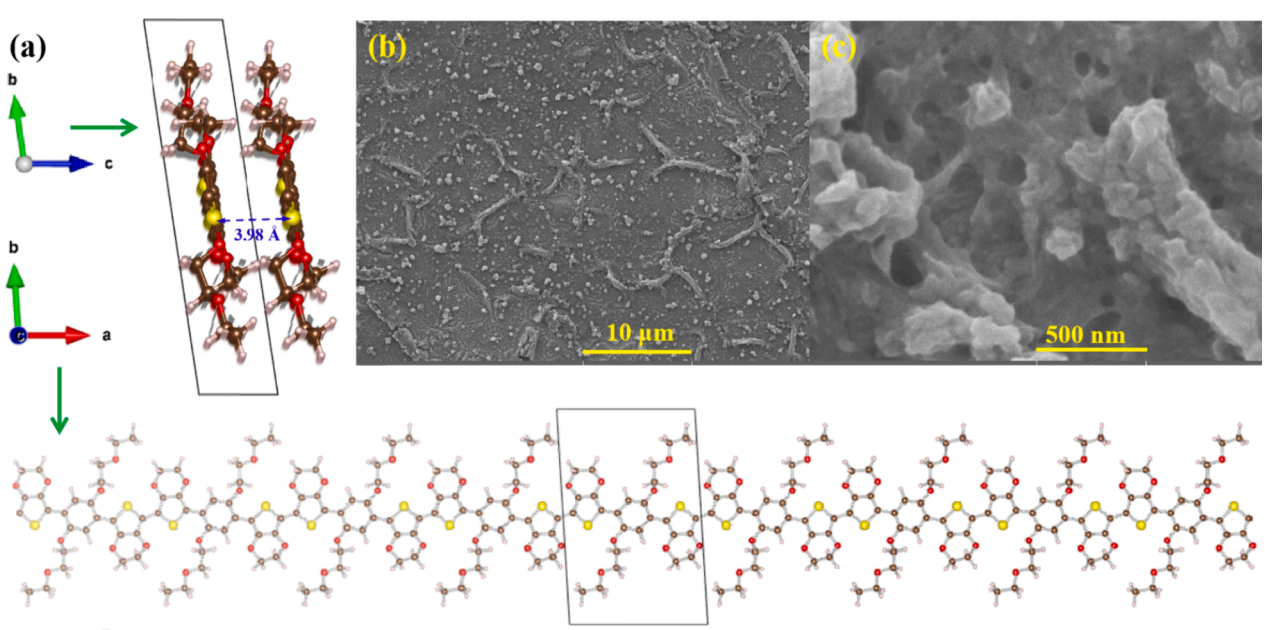


Fig. 2. (a) Optimized stacking structure of poly(EB-ether): side view and top view; morphology of poly(EB-ether) film at magnifications of ×10000 (b) and ×200000 (c).

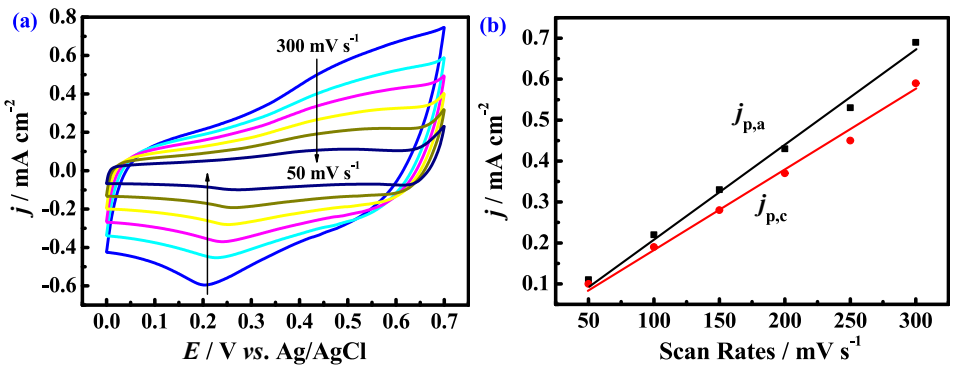


Fig. 3. (a) Cyclic voltammograms of poly(EB-ether) in H_2O-KCl (1 mol/L) at different scan rates; (b) linear relationship of poly(EB-ether) between current densities of redox peaks and scan rates, $j_{p,a}$: current density of anodic peak; $j_{p,c}$: current density of cathodic peak.

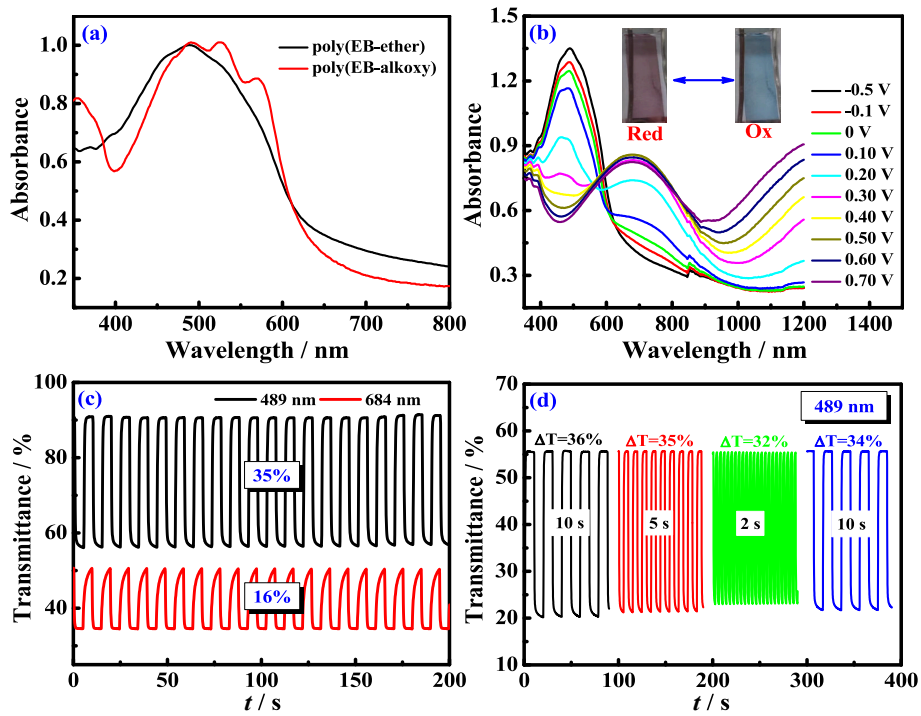


Fig. 4. (a) Optical absorption spectra of poly(EB-ether) and poly(EB-alkxoy) at neutral state; (b) spectroelectrochemistry of poly(EB-ether) in H_2O-KCl (1.0 mol/L); (c) time-transmittance curves of poly(EB-ether) in H_2O-KCl (1.0 mol/L), switching time: 10 s; (d) time-transmittance curves of poly(EB-ether) in H_2O-KCl (1.0 mol/L) with different switching times.

When applied potential turns from −0.5 V to 0.7 V, the absorption peak at 489 nm disappears gradually and two new absorption peaks at 684 nm and >1000 nm appear (Fig. 4b). The optical absorption change of poly(EB-ether) indicates polaron and bipolaron occur. In oxidation process, the appearance color of poly(EB-ether) turns from dark red (L^*: 39.8; a^*: 31.4; b^*: 18.4) to light blue (L^*: 52.7; a^*: −4.9; b^*: −11.6) gradually.

Also, the kinetic result of poly(EB-ether) was characterized by time-transmittance curves, as shown in Fig. 4c. And, the electrochromic parameters including optical contrast (ΔT), response time and coloration efficiency (CE) were calculated and summarized in Table 1. At the switching time of 10 s, the optical contrasts of poly(EB-ether) are 35 % at 489 nm and 16 % at 684 nm, respectively (Fig. 4c). And, the dependence of optical contrast on switching time was explored, as shown in Fig. 4d. It could be found that the optical contrast at 489 nm shows slight decrease as switching time turning from 10 s to 2 s. When the switching time turns to 10 s, the optical contrast becomes 34 %, which is lower than initial value (36 %). The phenomenon may be ascribed to the degradation of polymer. And, the kinetic stability of poly(EB-ether) was measured, as shown in Fig. S4. After 100 cycles, the optical contrast of poly(EB-ether) turns from 36 % to 34 %, and remains 94 % kinetic stability.

Table 1
The electrochromic parameters of poly(EB-ether) and poly(EB-alkoxy).

Compound	λ/ nm	ΔT	Response time / s		CE / C^{-1} cm^2		ref
			Red	Ox	Red	Ox	
poly(EB-alkoxy)	469	29 %	2.5	0.8	226	209	[35]
	641	17 %	3.3	0.4	169	157	
poly(EB-ether)	489	35 %	1.2	0.8	283	205	This work
	684	16 %	3.7	0.8	162	118	

λ: wavelength; Ox: oxidation process; Red: reduction process.

In oxidation process, all the response times of poly(EB-ether) were calculated to be 0.8 s at 489 nm and 684 nm; In reduction process, the

response times were 1.2 s at 489 nm and 3.7 s at 684 nm, respectively (Fig. 5a~5b). It could be found that the response times in oxidation process are faster than those in reduction process. The phenomenon may be ascribed to following mechanism. Under the potential of 0.7 V, the polymer is oxidized gradually, meanwhile the doping ions move in polymer. During the process, its charge transport ability increases gradually [36–38]. Under driving force of 0.7 V, the increased charge transport ability further enhances doping ions move in polymer. In contrast, the charge transport ability of polymer decreases gradually during reduction process. Due to the reduced charge transport ability, the potential of −0.5 V provides little driving force for doping ion moving in later reduction stage. Also, the coloration efficiency were calculated according to time-transmittance curves and time–current curves (Fig. 5c~5d), which is a parameter to measure the energy consumption of electrochromic material operating. During oxidation process, *CE* are 205 C^{-1} cm^2 at 489 nm and 118 C^{-1} cm^2 at 684 nm; during reduction process, *CE* are 283 C^{-1} cm^2 at 489 nm and 162 C^{-1} cm^2 at 684 nm, respectively.

In short, poly(EB-ether) prepared in EtOH-CH_2Cl_2 (v/v = 5/1) mixed solvent possesses similar electrochromic properties with poly(EB-alkoxy), as shown in Table 1. And, poly(EB-ether) exhibits higher optical contrast at 489 nm relative to poly(EB-alkoxy). These results indicate EtOH-CH_2Cl_2 (v/v = 5/1) mixed solvent has no negative influence on electrochromic properties of poly(EB-ether), which further prove that the introduction of ether in precursor is a feasible method to decrease or avoid the use of halogenated solvent in electrochemical polymerization process of precursor.

2.6. Construction and properties of electrochromic device

Finally, the electrochromic device was constructed, which contain poly(EB-ether) modified ITO glass as working electrode, free ITO glass as counter electrode, and gel electrolyte (Fig. 6a). Firstly, the electrochromic performance of device was measured, as shown in Fig. 6b. The neutral poly(EB-ether) presents transmittance of 32 % at 450 nm and 53 % at 665 nm, while the oxidized poly(EB-ether) presents transmittance of 40 % at 450 nm and 49 % at 665 nm. The optical contrast of device is lower than pure poly(EB-ether) film, which is due to the different electrolyte used in device. From the video S1, the device could achieve reversible color change during redox process. And, the optical stability of device was also measured, as shown Fig. 6c. After 500 cycles, the optical contrasts of device at different wavelengths were all slightly decreased.

3. Conclusion

To decrease the use of halogenated solvent, an ether chain modified precursor EB-ether was prepared and electrochemical polymerized to corresponding polymer poly(EB-ether) in EtOH-CH_2Cl_2 (v/v = 5/1) containing 0.1 M Bu_4NPF_6 at low potential. Poly(EB-ether) could achieve optical contrast of 35 %, response time of 0.8 s, coloration efficiency of 283C^{-1} cm^2, and reversible color change between dark red and blue. These electrochromic parameters are similar with these of poly(EB-alkoxy) with alkoxy groups, which indicate that the introduction of ether chain in precursor is a feasible method to decrease or avoid the use of halogenated solvent in electrochemical polymerization process of precursor. Finally, the electrochromic device was constructed, which could achieve reversible color change during redox process and keep stable after 500 cycles.

CRediT authorship contribution statement

Shouli Ming: Writing – original draft, Investigation, Funding acquisition. **Zengshen Li:** Investigation, Formal analysis. **Hui Sun:** Resources, Investigation. **Ruyue Du:** Investigation. **Jilin Zhang:** Investigation. **Jinsheng Zhao:** Supervision, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

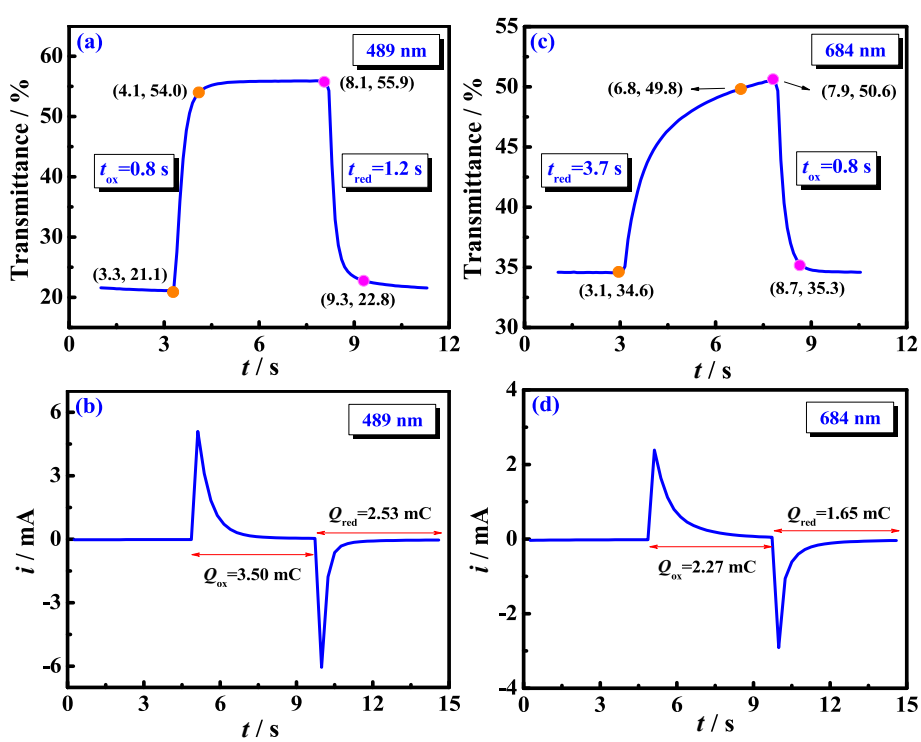


Fig. 5. Response times of poly(EB-ether) at 489 nm (a) and 684 nm (b); time–current curves of poly(EB-ether) at 489 nm (c) and 684 nm (d).

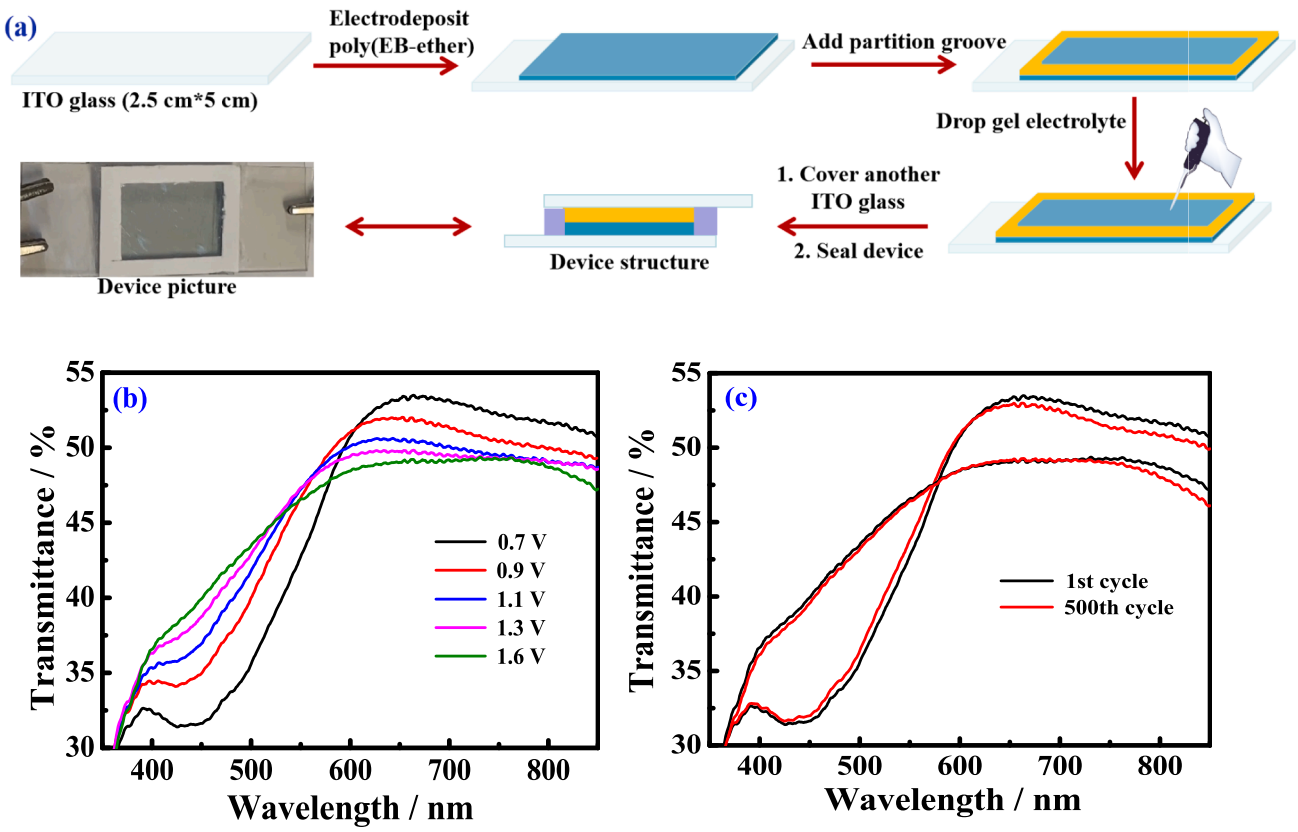


Fig. 6. Preparation process (a), electrochromic performance (b) and stability (c) of electrochromic device.

Data availability

Data will be made available on request.

Acknowledgement

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113364.

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